Water Quality Analysis in Indian Rivers Using PySpark and Machine Learning

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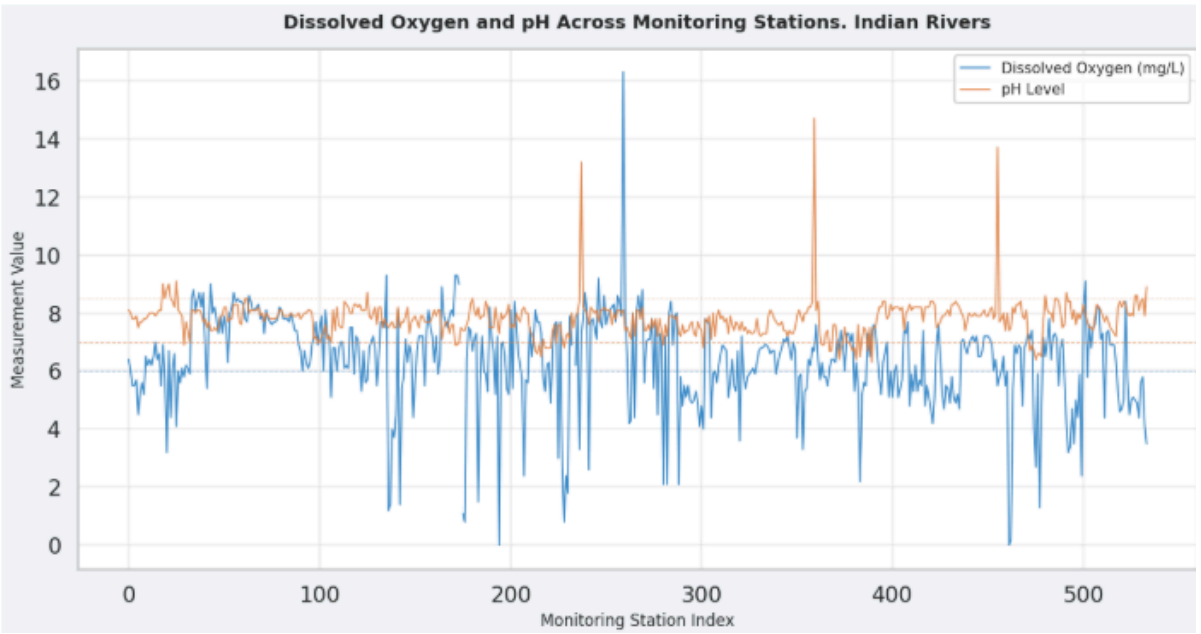
Water Quality in India: What the Data is Telling Us

This report goes through the water quality data collected from rivers across 18 states in India. The dataset has 534 measurement stations and looks at things like dissolved oxygen, pH, bacteria levels, and other indicators that together tell us how clean or dirty the water actually is.

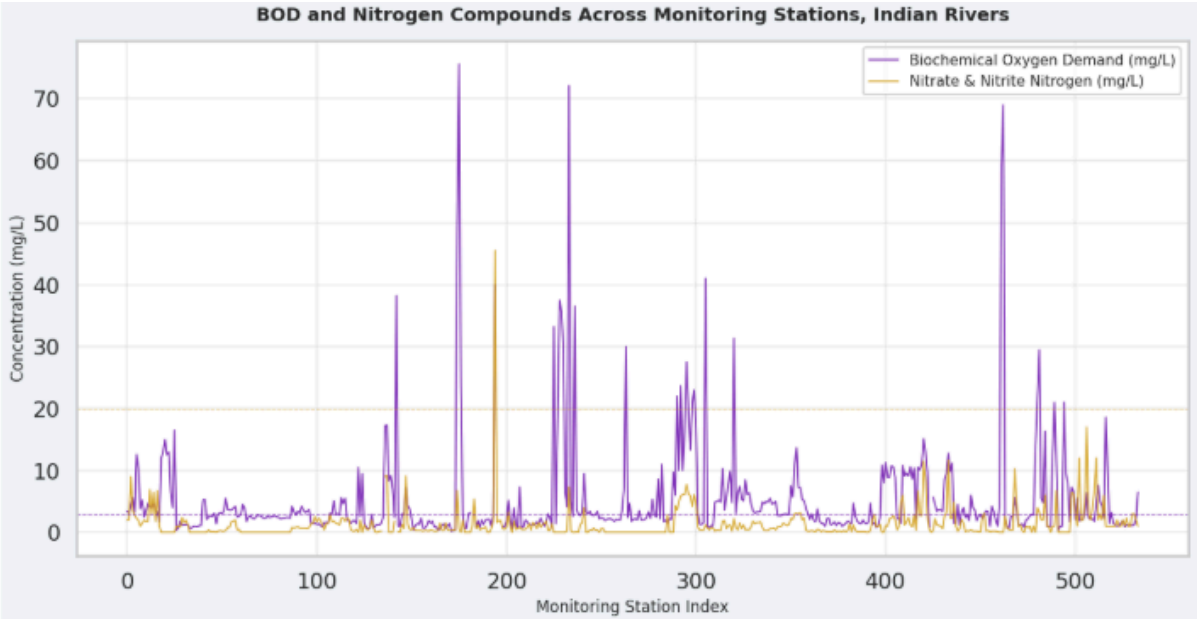
What we looked at

The data comes from official monitoring stations spread across India. Each station measured a handful of things: temperature, dissolved oxygen, pH, conductivity, organic demand, nitrogen compounds, and fecal bacteria. We took all of that and combined it into a single score called WQI, which stands for Water Quality Index. The higher the score, the worse the water. Simple as that.

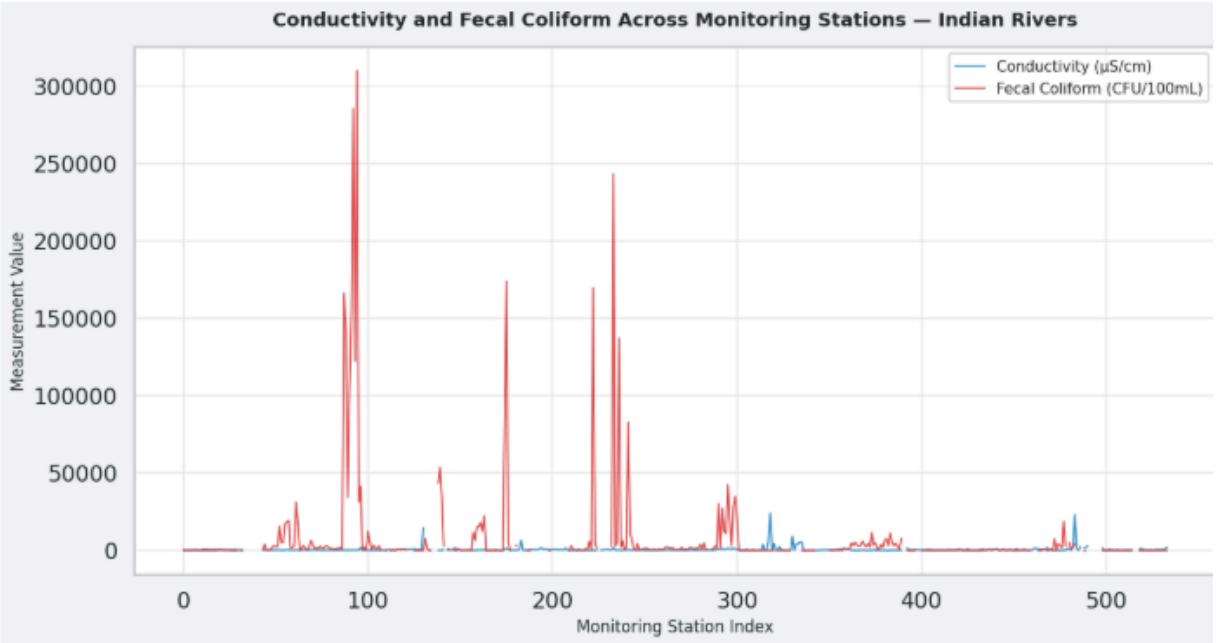
The charts and what they show



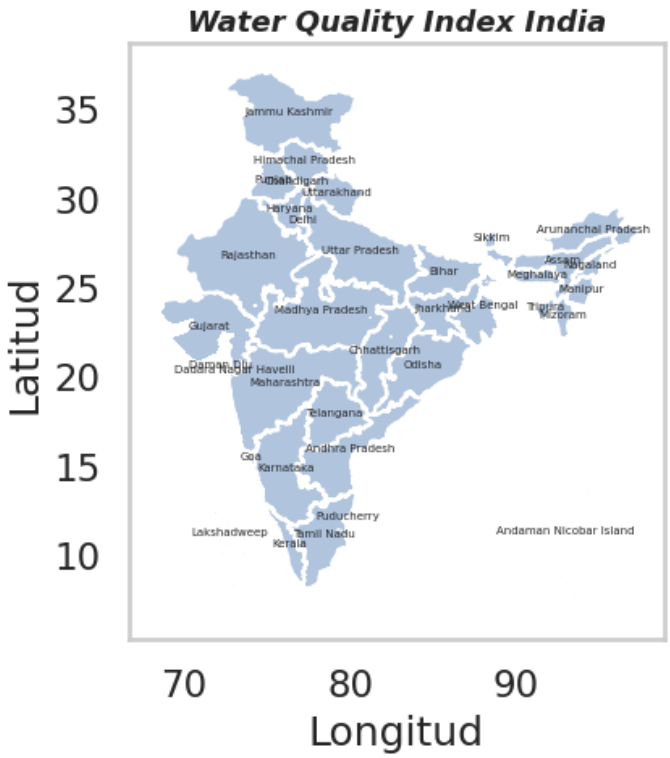
pH stays stable around 8 across almost all stations, which means the water is slightly alkaline but still within normal range. Dissolved oxygen is much more unstable and drops below 6 mg/L in several stretches, which is the minimum threshold to support healthy aquatic life. The extreme spikes visible in both variables around stations 260 and 450 suggest point source contamination or unusual environmental conditions in those specific areas.

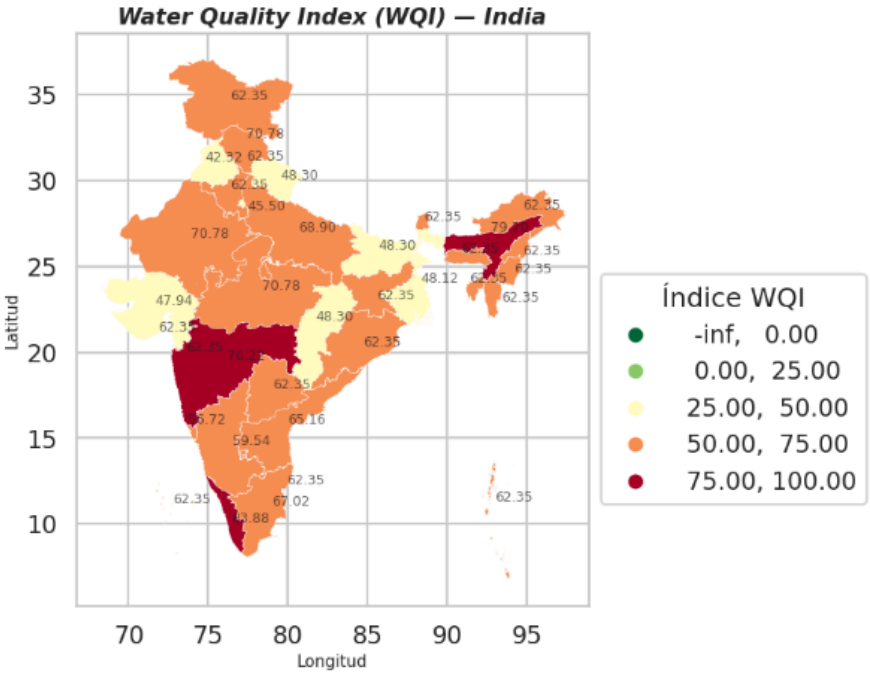


BOD stays low for most stations, which is a good sign, but there are sharp spikes reaching above 70 mg/L in a few locations. Those spikes point directly to heavy organic pollution, likely from sewage or industrial discharge near those stations. Nitrogen levels are generally flat and low across the dataset, but there is one notable spike around station 200 that goes up to 45 mg/L, which is high enough to cause algae overgrowth in the river. Both variables behave well in most of the country but the outliers are severe enough to be a real concern.

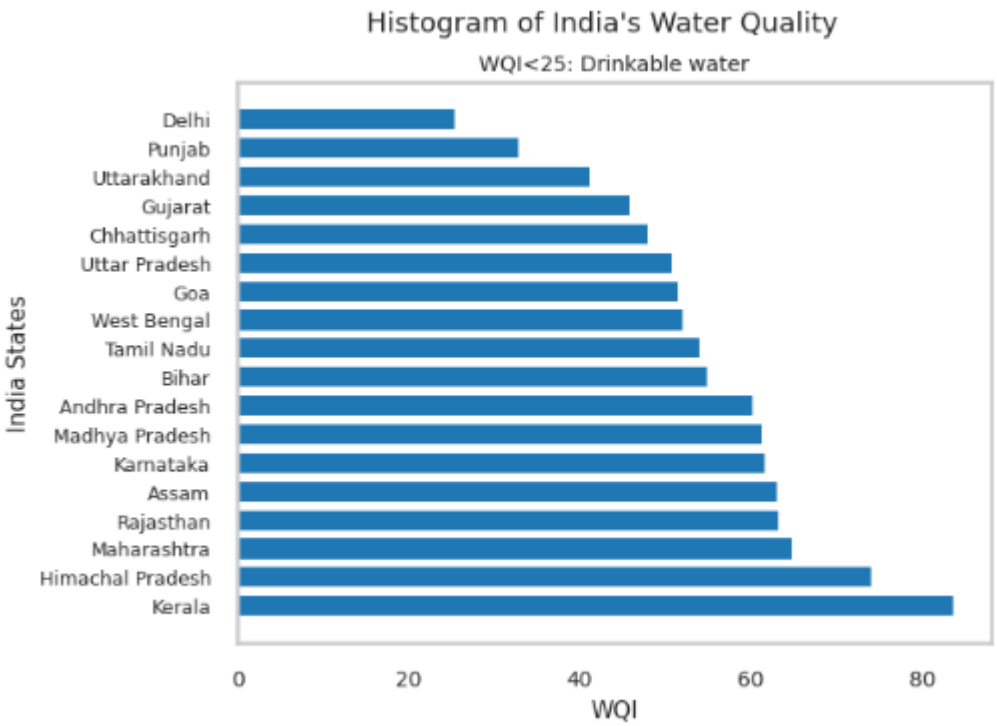


Conductivity stays mostly flat and low across all stations, meaning the mineral content of the water is generally manageable throughout the country. Fecal coliform is a completely different story. Most stations are near zero but there are several violent spikes, with one reaching above 300000 CFU per 100mL around station 90 and another close to 250000 around station 220. Those numbers are catastrophically high. Safe drinking water should be at zero and even safe recreational water should stay below 200. Whatever is happening at those specific stations points to severe sewage contamination with no treatment whatsoever.





The map of India colored by WQI score tells the clearest story. Most of the country falls in the medium to high range, meaning the water is hard and has notable mineral or biological contamination. The northern states, especially Punjab, come out looking relatively better. The southern states like Kerala and Karnataka tend to have higher WQI scores, not necessarily because the water is more polluted, but because the mineral content is naturally higher in those regions.



The bar chart ranking each state by average WQI confirms this. Punjab sits at the top as the best performer. A bunch of states cluster in the middle with similar scores. Delhi, Uttarakhand, and Chhattisgarh show up with lower scores but that is partly because those stations had missing data, so the numbers are not fully reliable for those places.

What the model learned

We trained a neural network to predict the WQI score based on the six quality indicators. The training loss chart shows the model figured out the basic patterns very fast, within the first 10 or so training rounds the error dropped dramatically and then stayed flat. That is actually a good thing because it means the model was not jumping all over the place.

The scatter plot comparing real WQI values versus what the model predicted shows most points clustering near the diagonal line, which is the perfect prediction line. That means in most cases the model got pretty close to the actual value. There are some outliers where it missed, which is expected with a dataset this size.

What this is actually telling us in plain terms

If you had to pick the cleanest water in this dataset, Punjab is your best bet. If you are looking at the worst, stations in Assam, West Bengal, and parts of Maharashtra show really high bacteria counts that are a serious concern for public health.

The fecal coliform numbers at some stations are extreme. We are talking values in the hundreds of thousands. That is not water you want anywhere near drinking or even recreational use. The good news is that most of the high WQI scores are driven by mineral hardness, not contamination, which is a very different kind of problem and much less immediately dangerous.

Temperature across the country stays in a range that is not alarming on its own. Dissolved oxygen at the lower-scoring stations is the bigger concern because low oxygen in a river means the ecosystem is struggling, which usually signals organic pollution upstream.

What could happen next based on the patterns

Based on what the model learned, a few things are predictable. Stations with high BOD and high fecal coliform almost always end up with a bad WQI score. So if you see those two going up in a region, the overall water quality is going to drop too. That is not a coincidence, it is a pattern the model picked up clearly.

States that are doing better right now like Punjab and parts of Gujarat could see their scores get worse if agricultural runoff increases, since nitrogen and nitrate levels are the main thing tipping those regions toward a worse category.

For the states already in the worse categories, the bacteria levels are the main thing to address. Industrial and urban discharge near the monitoring stations is the likely culprit, and without intervention those numbers are not going to improve on their own.

The model is not perfect and the dataset is relatively small for making national-level predictions, but the directional signals are clear. The water quality divide in India is real, it is geographic, and it is driven by a mix of natural mineral content and human-caused contamination depending on the region.

Conclusion

The data shows that clean water in India is not evenly distributed. Some states are doing okay, most are in a middle ground that needs attention, and a few have stations showing levels that are genuinely alarming. The good news is that the patterns are consistent enough that targeted interventions in the right places could make a measurable difference

References

- World Health Organization. Guidelines for Drinking-water Quality. 4th edition. Geneva: WHO Press, 2011.
- Han et al. Data Mining: Concepts and Techniques. 3rd edition. Morgan Kaufmann, 2011.